FOR EDITH · PAEDIATRICS & CHILD HEALTH JUNIOR CLERKSHIP

# Acute Asthma

Tutorial | Week 1 · Breathing, fever & the clinical methods | Paediatrics | Dr Mawanda

An acute asthma attack in a child is graded in the first two minutes (can the child talk or feed, $SpO_2$, respiratory rate, pulse, chest sounds), and treated at once with oxygen to $SpO_2$ 94–98%, repeated salbutamol by spacer or nebuliser, early oral or IV steroid, and escalation to ipratropium, IV magnesium and then aminophylline for severe or life-threatening attacks; no child goes home without a written plan, a spacer, checked inhaler technique and a controller inhaler.

**Must know before the ward round**

1. Severity (UCG 2023/BTS): SEVERE = can't complete sentences or too breathless to talk/feed, $SpO_2$ <92%, HR >140 (2–5 y) or >125 (>5 y), RR >40 (2–5 y) or >30 (>5 y), PEF 33–50%. LIFE-THREATENING = silent chest, cyanosis, poor effort, exhaustion, drowsiness/confusion, hypotension, PEF <33%.
2. A quiet child with a silent chest and falling respiratory rate is tiring, not improving.
3. Oxygen to keep $SpO_2$ 94–98% (GINA 2025); UCG 2023 says high-flow oxygen to keep $SpO_2$ ≥94%.
4. Salbutamol: pMDI 100 µg/puff via spacer 2–10 puffs (one puff at a time, 5 breaths each), or nebulised 2.5 mg (<5 y; UCG 2023) / 2.5–5 mg (≥5 y), every 20 minutes for the first hour, then space out by response.
5. Ipratropium 250 µg nebulised (children <12 y) every 20–30 min for the first 2 hours in severe/life-threatening attacks, mixed with salbutamol (UCG 2023).
6. Steroid for every attack needing more than one dose of salbutamol: prednisolone 1–2 mg/kg once daily (max 20 mg <2 y, 30 mg 2–5 y, 40 mg 6–11 y) for 3 days; or hydrocortisone 4 mg/kg IV 6-hourly (max 100 mg) if vomiting or too ill to swallow.
7. IV magnesium sulfate 40–50 mg/kg (max 2 g) over 20 minutes is the first IV bronchodilator for severe attacks not responding to inhaled therapy (GINA 2025, BTS/SIGN, RCH).
8. Aminophylline 5 mg/kg IV over 20 minutes (omit the load if on oral theophylline), then 1 mg/kg/h; UCG 2023 still lists it but discourages it because of toxicity.
9. Never sedate an asthmatic child, never give routine antibiotics (fever = think pneumonia), and never give a β-blocker.
10. Discharge when stable on salbutamol 3–4-hourly, $SpO_2$ ≥94% in air, PEF ≥60–80% of best, inhaler technique checked, controller started or stepped up, written action plan, review in 2 days to 1 week.
11. Chronic control (GINA 2025): 6–11 y daily low-dose ICS from Step 2; ≥12 y as-needed ICS-formoterol (AIR) is preferred; ≤5 y daily low-dose ICS if attacks are frequent or severe. A home-made 500 mL bottle spacer works as well as a commercial one.
12. Most childhood attacks are triggered by viral infections; smoke from charcoal/firewood, dust and cold are the common Ugandan triggers.


1 In a nutshell
2 Definitions & classification
3 Epidemiology & Uganda context
4 Pathophysiology
5 Clinical acumen
6 Clinical workup
7 Differential diagnosis
8 Investigations
9 Management
10 Complications
11 Ward-round & exam pearls
12 Mnemonics & memory aids

# In a nutshell

**Acute asthma** (an "asthma attack" or "exacerbation") is a sudden or gradual worsening of breathlessness, cough, wheeze or chest tightness in a child with asthma, caused by **bronchospasm, airway wall oedema and mucus plugging** in the small airways. In Uganda it is one of the commonest reasons a school-age child is carried into the emergency room at night, and children still die from it, usually because the attack was **under-graded**, treated with too little salbutamol, given oxygen late, or sent home without a plan.

The whole tutorial rests on three moves:

1. **Grade the attack in two minutes**: talk/feed, $SpO_2$, respiratory rate, pulse, chest sounds, conscious level. Mild–moderate, severe or life-threatening.
2. **Treat by grade, immediately**: oxygen to 94–98%, salbutamol every 20 minutes for the first hour, steroid early; add ipratropium, then **IV magnesium**, then aminophylline, and call for help when the child is not responding.
3. **Prevent the next attack**: inhaler technique with a spacer, a controller (inhaled corticosteroid), trigger avoidance, a written action plan and follow-up.

> **CLINICAL PEARL**
> The one thing never to forget: **a quiet chest is not a better chest.** A child who stops wheezing, becomes drowsy and breathes more slowly while $SpO_2$ falls is about to arrest. That is a **silent chest**: the airways are so narrow that no air moves. Treat as life-threatening and call the most senior person.

# Definitions & classification

| Term | Meaning |
|---|---|
| **Asthma** | Chronic inflammatory airway disease with **variable** respiratory symptoms (wheeze, cough, breathlessness, chest tightness) and **variable expiratory airflow limitation** (GINA 2025) |
| **Acute asthma / exacerbation** | Progressive worsening of symptoms and lung function beyond the child's usual day-to-day variation, needing a change in treatment |
| **Wheeze** | Continuous, high-pitched, mainly **expiratory** musical sound from narrowed intrathoracic airways (compare **stridor**, an inspiratory sound from the larynx/trachea) |
| **Preschool wheeze** | Wheeze in children <5 years. Two patterns: **episodic viral wheeze** (only with colds, well in between) and **multiple-trigger wheeze** (also with exercise, laughing, smoke, at night); the second is more likely to be asthma |
| **Status asthmaticus** | Older term for a severe attack that does not respond to repeated standard bronchodilator therapy; now called **acute severe** or **life-threatening asthma** |
| **Near-fatal asthma** | An attack needing ventilation or with $PaCO_2$ rising |

### Severity grading of the acute attack (UCG 2023, adapted from BTS/SIGN and GINA)

| Feature | Mild–moderate | Severe | Life-threatening |
|---|---|---|---|
| Talking / feeding | Talks in sentences; feeds | **Cannot complete sentences in one breath**; too breathless to talk or feed | Unable to talk; exhausted |

| Feature | Mild–moderate | Severe | Life-threatening |
|---|---|---|---|
| Behaviour | Alert, may be agitated | Agitated, sits hunched forward | **Drowsy, confused**, floppy |
| $SpO_2$ in air | **≥92%** | **<92%** | <92%, often much lower; **cyanosis** |
| Respiratory rate | ≤40 (2–5 y), ≤30 (>5 y) | **>40 (2–5 y); >30 (>5 y)** | May be **falling** (exhaustion), poor effort |
| Pulse | ≤140 (2–5 y), ≤125 (>5 y) | **>140 (2–5 y); >125 (>5 y)** | **Bradycardia**, arrhythmia, **hypotension** |
| Accessory muscles | Not used or mild | **Used** (sternomastoid, subcostal/intercostal recession, nasal flaring) | Exhausted, paradoxical breathing |
| Chest sounds | Wheeze | Loud wheeze (both phases) | **Silent chest** |
| PEF (≥5–6 y who can do it) | **>50%** predicted/best | **33–50%** | **<33%** |

Any **one** feature of a higher grade puts the child in that grade. Not all features need to be present. **If the parent or child says this is the worst attack ever, believe them.**

### Severity of the underlying asthma (for the discharge plan)

| GINA 2025 term | What it means |
|---|---|
| **Well controlled** (in the last 4 weeks) | Daytime symptoms ≤2/week (≤1/week for ≤5 y), no night waking, reliever ≤2/week, no activity limitation |
| **Partly controlled** | 1–2 of the above not met |
| **Uncontrolled** | 3–4 not met |
| **Risk factors for a fatal attack** | Previous ICU/intubation; ≥2 admissions or ≥3 ER visits in the last year; using >1 salbutamol canister a month; not on ICS; recent oral steroids; poor adherence; psychosocial problems; food allergy; poor perception of severity |

UCG 2023 still uses the older "intermittent / mild persistent / moderate persistent / severe persistent" steps for chronic treatment (see Management). Know both vocabularies.

## Epidemiology & Uganda context

- Asthma is **the most common chronic disease of childhood** worldwide (WHO fact sheet, 2026); it affected an estimated **363 million people in 2023** and caused about 442,000 deaths, mostly in low- and middle-income countries.
- In Ugandan schools surveys have found wheeze in roughly **one in ten** children, more in urban Kampala than rural areas, and most are **undiagnosed or using only salbutamol tablets or syrup**.
- **Triggers you will hear about in Kampala**: viral colds (the commonest), **smoke** from charcoal stoves and firewood, kerosene lamps, cigarette smoke, dust (house dust mites, dusty roads, dry season), cold early mornings, cockroaches, pets, exercise, strong perfumes, aspirin/NSAIDs.
- **What is available**:

| Level | Usually available | Often missing |
|---|---|---|
| **HC III** | Salbutamol tablets and inhaler, prednisolone tablets, sometimes a nebuliser, adrenaline | Oxygen, pulse oximeter, spacer |

| Level | Usually available | Often missing |
|---|---|---|
| **HC IV / general hospital** | Oxygen concentrator, pulse oximeter, nebuliser, salbutamol nebules, hydrocortisone, aminophylline, beclomethasone inhaler | Ipratropium, IV magnesium (stocked for eclampsia in maternity!), peak flow meter |
| **Regional / national referral (Mulago, Mengo, Lubaga, Nsambya)** | All the above, ipratropium, $MgSO_4$, blood gas (some), HDU, CXR | PICU beds are very few (Mulago, some private hospitals) |

UGANDA CONTEXT

**Magnesium sulfate is on every Ugandan labour ward** (50% ampoules for eclampsia), so it is usually available even when the paediatric ward has none. Borrow it, but **calculate the paediatric dose carefully** (40–50 mg/kg, max 2 g) and dilute it: the eclampsia loading dose (4 g IV) would be a dangerous overdose in a child.

UGANDA CONTEXT

Many Ugandan children are still treated with **oral salbutamol tablets or syrup** and **oral aminophylline**. They work slowly, cause tremor and tachycardia, and are no substitute for an inhaler with a spacer. A **500 mL plastic water bottle spacer** (a hole cut in the base to fit the inhaler mouthpiece, sealed with tape) is free and works.

# Pathophysiology

## Why the airway narrows

1. **Chronic eosinophilic inflammation** (Th2 cytokines IL-4, IL-5, IL-13; mast cells, IgE) makes the airway **hyper-responsive**.
2. A trigger (usually a **rhinovirus** cold) causes:
   - **Bronchial smooth-muscle spasm** (reverses quickly with **$\beta_2$-agonists**; ipratropium blocks vagal tone),
   - **Mucosal oedema** and inflammatory cell influx (reversed over **hours** by **corticosteroids**),
   - **Mucus plugging** (the reason severe attacks respond slowly and can cause atelectasis).
3. Narrowing is worse in **expiration** (airways are compressed when intrathoracic pressure rises), so air is trapped: **hyperinflation**.

## Why children decompensate quickly

- Small airways: resistance is proportional to $1/\text{radius}^4$; a little oedema narrows a child's airway far more than an adult's.
- A compliant chest wall: the ribs are drawn in (recession) rather than lifting the chest, so work of breathing is wasted.
- Fewer collateral channels and a diaphragm that fatigues earlier.

## Blood gases through an attack

| Phase | $PaO_2$ / $SpO_2$ | $PaCO_2$ | Meaning |
|---|---|---|---|
| Early | Normal or slightly low | **Low** (hyperventilation) | Expected |
| Worsening | Falling (V/Q mismatch) | **Normal** | **Danger**: a "normal" $CO_2$ in a breathless child means tiring |

| Phase | $PaO_2$ / $SpO_2$ | $PaCO_2$ | Meaning |
|---|---|---|---|
| Life-threatening | Low | **Rising** (>6 kPa / 45 mmHg) | Respiratory failure: needs HDU/ICU and ventilation |

### Why the drugs work (and their side effects)

| Drug | Mechanism | Main side effects |
|---|---|---|
| **Salbutamol** | $\beta_2$-agonist: ↑cAMP, relaxes airway smooth muscle in minutes | Tachycardia, tremor, **hypokalaemia**, raised **lactate**, hyperglycaemia, restlessness |
| **Ipratropium** | Antimuscarinic: blocks vagal bronchoconstriction | Dry mouth; additive to salbutamol in the first hours only |
| **Corticosteroids** | Switch off inflammatory gene transcription, ↓oedema; restore β-receptor response | Effect from **3–4 h**; hyperglycaemia, mood change; short courses are safe |
| **Magnesium sulfate** | Blocks calcium channels in smooth muscle → bronchodilation | Flushing, **hypotension**, weakness, loss of reflexes at high levels |
| **Aminophylline** | Phosphodiesterase inhibitor, adenosine antagonist | **Narrow therapeutic window**: vomiting, tachyarrhythmia, **seizures** |
| **Oxygen** | Corrects V/Q mismatch hypoxaemia | None in asthma at these doses (unlike COPD) |

CLINICAL PEARL

The salbutamol paradox: a child who has had hourly nebulised salbutamol can be **tachycardic, tremulous, anxious, with a high lactate and a low potassium**. These are drug effects, not necessarily a worse attack. Look at the **work of breathing and $SpO_2$** to judge the asthma, and check $K^+$ if you have been giving salbutamol continuously.

# Clinical acumen

### Focused history

Take this **while the first salbutamol is running**:

- **Is this asthma?** Previous episodes of wheeze, night cough, exercise cough, relief with inhalers, atopy (eczema, allergic rhinitis), family history. A first-ever wheeze in an infant is more often **bronchiolitis**.
- **This attack**: when it started, trigger (cold, smoke, exercise, food, drug), treatment already taken (how many puffs, with a spacer? tablets?), how they responded.
- **Danger history**: previous **ICU or intubation**, admissions or emergency visits in the past year, oral steroid courses, how much salbutamol they use per month.
- **Usual treatment and adherence**: is there an inhaled steroid at home? Is it taken? Who gives it? Does the child have a spacer?
- **Fever, chest pain, foul sputum** (pneumonia), **sudden onset while eating or playing** (foreign body), **rash, lip swelling, food or sting** (anaphylaxis).
- **Background**: HIV status (and cotrimoxazole/ART), TB contact, growth, immunisations (UNEPI, including PCV and measles), home fuel (charcoal, firewood, kerosene), smokers, school, cost of inhalers.

### Examination

1. **Look first, touch later**: posture (tripod, sitting up), ability to talk or feed, colour, alertness, sweating.
2. **Count** respiratory rate for a full minute; heart rate; **$SpO_2$ in air** (if safe) then on oxygen.

3. **Work of breathing**: nasal flaring, tracheal tug, subcostal/intercostal recession, sternomastoid use, abdominal breathing; **pulsus paradoxus** is no longer used routinely.
4. **Chest**: hyperinflation, air entry on both sides, wheeze (expiratory, both phases, or **silent**), **crackles or bronchial breathing** (pneumonia, collapse), **unequal air entry** (pneumothorax, foreign body, mucus plug).
5. **Look for complications**: subcutaneous emphysema (crepitus in the neck), tracheal deviation.
6. **Rest of the child**: temperature, hydration, eczema, nasal polyps (think cystic fibrosis), clubbing (asthma does **not** cause clubbing: think bronchiectasis, CF, TB), growth (weight-for-age and height Z-scores), BCG scar.

### Red flags

RED FLAG

**Life-threatening asthma, act now and call the most senior person:** silent chest, cyanosis, $SpO_2$ <92% despite oxygen, poor respiratory effort, exhaustion, **drowsiness or confusion**, bradycardia or hypotension, PEF <33%, a normal or rising $CO_2$. Also: any child who has needed salbutamol every 20 minutes for an hour and is not better.

RED FLAG

**Not asthma, or asthma plus something:** unilateral wheeze or reduced air entry (foreign body, pneumothorax); fever with focal crackles (pneumonia); stridor (upper airway); urticaria, facial swelling and hypotension (**anaphylaxis: IM adrenaline first**); a heart murmur, hepatomegaly and crackles (heart failure); chronic cough, weight loss and a TB contact.

### Thinking like a clinician

- **Grade, then treat, then re-grade every 20–30 minutes.** The grade changes, and so must the plan.
- **"All that wheezes is not asthma."** Under 2 years, think bronchiolitis first. Sudden onset with choking = foreign body until proven otherwise. **Unilateral** findings are never asthma alone.
- **Fever is not part of asthma.** UCG 2023: "if fever present, refer to pneumonia". Viral triggers cause low-grade fever, but a high fever with focal signs means look for pneumonia and treat it.
- **Don't be reassured by a falling respiratory rate** unless the child also looks better and $SpO_2$ is rising.
- **Salbutamol tablets at home do not count as treatment.** A child "on treatment" may never have had an inhaled drug.
- **Agitation is hypoxia until proven otherwise**: give oxygen, never diazepam or promethazine.
- **Anaphylaxis can look like asthma.** Wheeze + urticaria/angioedema + hypotension after a food, drug or sting → **IM adrenaline 0.01 mg/kg (0.01 mL/kg of 1:1000), max 0.5 mg**, into the outer thigh, before anything else.
- **Every admission is a failure of prevention.** Ask yourself: why did this child get this bad? (No ICS? No spacer? Charcoal stove? Cost?)

## Clinical workup

### Step 0: First 5 minutes

1. **ETAT triage**: respiratory distress = **emergency sign** if severe (unable to talk/feed, central cyanosis, $SpO_2$ <90%, exhaustion). Move the child to the **resuscitation area**.
2. **Sit the child up** on the parent's lap; do not force them to lie down.
3. **Oxygen** by nasal prongs (1–2 L/min) or face mask (5–10 L/min, UCG: at least 5 L/min) to keep **$SpO_2$ 94–98%**.
4. **Grade the attack** using the table above in under 2 minutes.

5. **First salbutamol now**: spacer (2–10 puffs) or nebulised **2.5 mg** (<5 y) / **2.5–5 mg** (≥5 y), driven by oxygen at 6–8 L/min if severe.
6. **Severe or life-threatening**: add **ipratropium 250 µg** to the nebuliser; call for senior help; prepare IV access; **prednisolone** (or IV hydrocortisone if vomiting or drowsy).
7. **Weigh the child** (or estimate: (age + 4) × 2 kg for 1–10 years) so every subsequent drug is weight-based.

> **RED FLAG**
> If the child is **drowsy, cyanosed or has a silent chest**: life-threatening asthma. Back-to-back nebulised salbutamol + ipratropium on oxygen, IV access, **IV hydrocortisone**, **IV magnesium** within the first hour, and call the most senior clinician and anaesthetist. Prepare bag-valve-mask. If the child stops breathing, start paediatric life support.

## Step 1: Focused history

- [ ] **Age** → <2 years: bronchiolitis/viral wheeze more likely; ≥5–6 years: can do PEF.
- [ ] **Onset and trigger** → sudden choking = foreign body; food/sting/drug = anaphylaxis; cold = viral exacerbation.
- [ ] **Treatment already given** (number of puffs, spacer, nebuliser, steroid) → a child who had 10 puffs at home and is still severe is a **poor responder**.
- [ ] **Previous ICU, intubation, admissions, steroid courses** → high risk of a fatal attack: lower your threshold to escalate.
- [ ] **Controller use and adherence** → informs discharge plan.
- [ ] **Fever, focal pain, purulent sputum** → pneumonia (antibiotics).
- [ ] **Contacts with TB, weight loss** → TB can present with wheeze from lymph-node compression of a bronchus.
- [ ] **HIV status, cotrimoxazole, ART** → PCP, LIP (lymphoid interstitial pneumonitis) and TB in the differential; drug interactions (theophylline with rifampicin and some ARVs).
- [ ] **Immunisation and nutrition** → measles/pertussis; malnutrition changes fluids and risks.
- [ ] **Home environment** → charcoal/firewood smoke, smokers, dust: trigger control.

## Step 2: Focused examination

**General**: posture, talk/feed, colour, consciousness (AVPU), sweating, hydration, weight.

**Vital signs**: RR (full minute), HR, $SpO_2$ (in air if safe, then on $O_2$), temperature, BP if severe (hypotension is late and ominous), capillary refill.

**Respiratory**: recession (subcostal, intercostal, suprasternal), nasal flaring, accessory muscles, chest shape (hyperinflation, Harrison sulci in chronic asthma), tracheal position, air entry equal?, wheeze, crackles, silent areas.

**Other**: eczema, allergic shiners, nasal polyps, clubbing (not asthma), murmurs, hepatomegaly, growth.

| Document these positives | And these negatives |
|---|---|
| Talk/feed ability and AVPU | "No silent chest; air entry equal" |
| $SpO_2$ in air and on $O_2$; RR, HR | "No focal crackles, no bronchial breathing" |
| Recession and accessory muscle use | "No stridor; no urticaria or angioedema" |
| Response after each salbutamol (time-stamped) | "Trachea central; no surgical emphysema" |

## Step 3: Bedside tests

| Test | How | What it tells you |
|---|---|---|
| **Pulse oximetry** | Correct-size probe, warm finger or toe, wait for a good waveform | **<92% in air = severe**; guides oxygen (target 94–98%). May overestimate $SpO_2$ in dark skin: read with the clinical picture |
| **Peak expiratory flow** (≥5–6 y) | Best of 3 blows standing; compare with personal best or predicted for height | <50% severe, <33% life-threatening; repeat 15–30 min after each treatment |
| **Random blood sugar** | Glucometer | Hypoglycaemia in a child who has not fed; hyperglycaemia from salbutamol/steroids |
| **Malaria RDT** (fever) | Finger prick | Co-existing malaria is common; fever needs a cause |
| **Response to salbutamol** | Re-grade 15–20 min after each dose | The most important "test"; a good response supports asthma |

## Step 4: Laboratory & imaging

| Test | When | Expected / abnormal finding | How it changes management |
|---|---|---|---|
| **Chest X-ray** | **Not routine.** Only if: life-threatening attack, poor response, suspected pneumothorax, focal signs, fever, first episode with doubt about diagnosis, foreign body | Hyperinflation, peribronchial thickening (normal for asthma); **pneumothorax, pneumomediastinum, collapse/consolidation, foreign body** | Chest drain for tension/large pneumothorax; antibiotics for pneumonia; bronchoscopy for foreign body. Available at HC IV (some) and all hospitals |
| **Blood gas** (capillary or arterial) | Life-threatening asthma or poor response; HDU | Normal/rising **$pCO_2$ (>6 kPa / 45 mmHg)** with hypoxaemia = respiratory failure; lactic acidosis from salbutamol | HDU/ICU, consider ventilation. **Mulago/private ICUs only**; many RRHs have no gas machine |
| **Serum potassium** | Repeated or continuous salbutamol; on aminophylline | **Hypokalaemia** (<3.5 mmol/L) | Add potassium to IV fluids. RRH/NRH |
| **Full blood count** | Fever, suspected infection | Neutrophilia (pneumonia; but steroids also raise WBC), eosinophilia (atopy, worms) | Antibiotics only for clear bacterial infection |
| **Theophylline level** | On aminophylline infusion with vomiting, arrhythmia or seizures | Toxic level | Stop infusion. Rarely available in Uganda: watch clinically |
| **HIV test** | Status unknown | Positive | ART, cotrimoxazole; widen differential (LIP, PCP, TB) |
| **GeneXpert (sputum/gastric/stool)** | Persistent cough >2 weeks, weight loss, TB contact | MTB detected | TB treatment (see the childhood TB topic) |

## Step 5: Putting it together

### Model one-line summary

> "A 7-year-old boy weighing 22 kg (weight-for-age Z-score −0.5), fully immunised, HIV-negative, known asthmatic on salbutamol inhaler only, with a 1-day history of worsening breathlessness after a cold. On arrival he could speak only in words, $SpO_2$ 89% in air, RR 38, HR 138, widespread wheeze, PEF 40% of

> predicted: **acute severe asthma** triggered by a viral URTI, in a child with poorly controlled asthma on no controller."

**Problem list**

1. Acute severe asthma, on treatment, response being assessed.
2. Hypoxaemia ($SpO_2$ 89% in air).
3. Poorly controlled chronic asthma: no ICS, no spacer, charcoal stove at home.
4. Viral URTI trigger (no evidence of pneumonia).

**Working diagnosis**: *acute severe exacerbation of asthma, viral trigger, uncontrolled background asthma.*

EXAM FAVOURITE

**Worked example 1: re-grading after treatment.** At 0 min: $SpO_2$ 89%, RR 38, HR 138, speaks in words, PEF 40% → **severe**. Given $O_2$, salbutamol 5 mg + ipratropium 250 µg nebulised ×3 over 1 h, prednisolone 40 mg (2 mg/kg would be 44 mg; max 40 mg). At 60 min: $SpO_2$ 95% on 2 L, RR 28, HR 150, speaks in sentences, PEF 65%, tremor. **Interpretation**: responding well; the higher HR and tremor are salbutamol effects. Plan: space salbutamol to 1–2-hourly, then 4-hourly; wean oxygen; continue prednisolone for 3 days; start ICS before discharge.

EXAM FAVOURITE

**Worked example 2: a "normal" blood gas that is not normal.** A 9-year-old girl after 2 hours of back-to-back nebulisers: drowsy, RR 22, quiet chest, $SpO_2$ 91% on mask. Capillary gas: pH 7.30, **$pCO_2$ 6.4 kPa (48 mmHg)**, lactate 3.5. **Interpretation**: a breathless asthmatic should be blowing off $CO_2$ (low $pCO_2$). A normal or high $pCO_2$ with drowsiness and a falling RR = **exhaustion and respiratory failure**. This is life-threatening asthma: IV magnesium now (if not given), consider aminophylline or IV salbutamol in HDU, and call the anaesthetist for possible intubation.

# Differential diagnosis

| Condition | Distinguishing features | Key investigation |
|---|---|---|
| **Bronchiolitis** | <2 years (peak 2–6 months), first episode, coryza then cough, **fine crackles** and wheeze, little response to salbutamol | Clinical; $SpO_2$ |
| **Pneumonia** | Fever, fast breathing, chest indrawing, **focal crackles/bronchial breathing**, no response to bronchodilators | CXR, FBC |
| **Foreign body aspiration** | Sudden choking while eating/playing (groundnuts, beans, maize), **unilateral** wheeze or reduced air entry, no atopy | Inspiratory and expiratory CXR; bronchoscopy |
| **Anaphylaxis** | Trigger (food, drug, sting), urticaria, angioedema, hypotension, vomiting | Clinical; **IM adrenaline** |
| **Croup / upper airway obstruction** | **Inspiratory stridor**, barking cough, hoarse voice | Clinical |
| **Heart failure** (VSD, RHD, cardiomyopathy) | Murmur, gallop, **hepatomegaly**, basal crackles, poor growth, sweating with feeds | CXR (cardiomegaly), echo |
| **Pulmonary TB / TB lymph nodes** | Persistent cough >2 weeks, weight loss, fever, TB contact; wheeze from node compression | Xpert, CXR (hilar nodes), TST |

| Condition | Distinguishing features | Key investigation |
|---|---|---|
| **HIV-related lung disease** (LIP, PCP, bronchiectasis) | HIV-positive, clubbing, parotid swelling (LIP), hypoxia out of proportion (PCP) | HIV test, CXR |
| **Recurrent aspiration / GORD** | Cough with feeds, cerebral palsy, vomiting | Clinical; swallow assessment |
| **Bronchiectasis, cystic fibrosis** | Chronic wet cough, **clubbing**, failure to thrive | CXR, sweat test (rarely available) |
| **Tropical pulmonary eosinophilia / Loeffler (ascaris larval migration)** | Wheeze, eosinophilia, worms | FBC (eosinophils), stool |

# Investigations

| Test | What you are looking for | Why |
|---|---|---|
| $SpO_2$ | <92% in air = severe | Guides oxygen and grade |
| PEF (≥5 y) | % predicted or best | Objective severity and response |
| CXR (selected) | Pneumothorax, consolidation, foreign body | Excludes complications and mimics |
| Blood gas (severe) | Rising $pCO_2$ | Detects respiratory failure |
| Potassium, glucose | Hypokalaemia, hyper/hypoglycaemia | Drug side effects |
| Spirometry with reversibility (when well, ≥5–6 y) | $FEV_1$/FVC <0.8–0.9; **$FEV_1$ rise ≥12%** after salbutamol (UCG 2023) | Confirms asthma diagnosis later in clinic |

# Management

## A. Principles

1. **Treat by grade and re-grade every 20–30 minutes.**
2. **Inhaled $\beta_2$-agonist is the backbone**; a spacer is as good as a nebuliser in mild–moderate and most severe attacks.
3. **Give a steroid early** (within the first hour) for any attack needing more than a single salbutamol dose.
4. **Escalate stepwise and quickly** when there is no response: ipratropium → IV magnesium → aminophylline (or IV salbutamol in HDU) → ventilation.
5. **No sedatives, no routine antibiotics, no cough syrups, no mucolytics, no chest physiotherapy** in the acute attack.
6. **Before discharge**: controller, technique, spacer, action plan, follow-up.

## B. Emergency algorithm (based on UCG 2023, GINA 2025, BTS/SIGN)

1. **Assess and grade** (2 minutes). Weigh.
2. **All grades**: sit up; **oxygen** if $SpO_2$ <94% (target **94–98%**).
3. **Salbutamol**:
   - **Mild–moderate**: pMDI 100 µg/puff via spacer, **2–10 puffs** (one puff at a time, 5 tidal breaths per puff), **every 20 minutes for 1 hour**; or nebulised 2.5 mg (<5 y) / 2.5–5 mg (≥5 y).
   - **Severe/life-threatening**: **nebulised salbutamol 2.5–5 mg + ipratropium 250 µg**, oxygen-driven (6–8 L/min), every 20 minutes (back-to-back if life-threatening).

4. **Steroid in the first hour**: prednisolone 1–2 mg/kg orally (max 20 mg <2 y, 30 mg 2–5 y, 40 mg ≥6 y), or **hydrocortisone 4 mg/kg IV** (max 100 mg) 6-hourly if vomiting, drowsy or life-threatening.
5. **Re-grade at 1 hour**:
   - **Good response** (talking, $SpO_2$ ≥94% in air, PEF >60–80%): space salbutamol to 1–4-hourly; observe 1–4 h; plan discharge.
   - **Incomplete response**: admit; continue salbutamol 1–2-hourly + ipratropium for the first 2 hours (then 4–6-hourly); steroid; oxygen.
   - **Poor response / life-threatening**: **IV magnesium sulfate 40–50 mg/kg (max 2 g) over 20 min**; senior review; HDU.
6. **Still not improving**: **aminophylline** 5 mg/kg IV over 20 min (omit if on oral theophylline), then 1 mg/kg/h infusion with cardiac monitoring; or **IV salbutamol** (HDU only, 15 µg/kg over 10 min, then infusion 1–5 µg/kg/min) where available.
7. **Respiratory failure** (exhaustion, rising $CO_2$, falling consciousness): anaesthetist/ICU; bag-valve-mask; intubation by the most experienced person (ketamine induction).
8. **Throughout**: monitor $SpO_2$, RR, HR, work of breathing every 15–30 min; $K^+$ and glucose if prolonged treatment; IV fluids at **maintenance only** if not drinking (no boluses unless shocked).

> **DRUG DOSING**
> **Never sedate** a breathless asthmatic child (diazepam, promethazine, phenobarbital, chloral). Restlessness is hypoxia. Sedation removes the drive that is keeping them alive.

## C. Drug table: acute asthma

| Drug | Dose | Route | Frequency | Duration / max | Notes |
|---|---|---|---|---|---|
| **Oxygen** | Nasal prongs 1–2 L/min (infants 0.5–1 L/min) or face mask 5–10 L/min | Inhaled | Continuous | To keep **$SpO_2$ 94–98%** | Use oxygen to drive the nebuliser in severe attacks (6–8 L/min) |
| **Salbutamol pMDI** | 100 µg/puff: **2–10 puffs** (GINA 6–11 y: 4–10 puffs) | Spacer (+ face mask if <3–4 y) | Every **20 min** in the first hour, then 1–4-hourly by response | Reduce to 4-hourly before discharge | 1 puff at a time, 5 breaths each; as effective as a nebuliser |
| **Salbutamol nebulised** | **2.5 mg** (<5 y); **2.5–5 mg** (≥5 y) in 2–4 mL saline | Nebuliser | Every 20 min ×3, then hourly to 4-hourly; back-to-back if life-threatening | – | UCG 2023: 5 mg adults, 2.5 mg children |
| **Ipratropium bromide** | **250 µg** (<12 y); 500 µg (≥12 y) | Nebulised with salbutamol | Every **20–30 min for the first 2 h**, then 4–6-hourly | Stop after 24–48 h | Severe and life-threatening only (UCG 2023) |
| **Prednisolone** | **1–2 mg/kg** once daily (UCG: 1 mg/kg; GINA: 1–2 mg/kg) | Oral | Once daily | **3 days** in children (UCG 2023; up to 5 days if slow); max **20 mg (<2 y), 30 mg (2–5 y), 40 mg (6–11 y)** | No taper needed for courses ≤14 days. Crush and mix with milk or juice |

| Drug | Dose | Route | Frequency | Duration / max | Notes |
|---|---|---|---|---|---|
| **Hydrocortisone** | **4 mg/kg** | IV | Every **6 h** | **Max 100 mg/dose**; switch to oral prednisolone when able | Use if vomiting, drowsy, life-threatening |
| **Dexamethasone** (alternative) | 0.3–0.6 mg/kg (max 16 mg) | Oral/IV | Once daily | 1–2 days | Used in some units for vomiting children; not in UCG for asthma |
| **Magnesium sulfate** | **40–50 mg/kg** (0.1 mL/kg of 50% = 50 mg/kg) | **IV** diluted (e.g. in 0.9% saline to ≤20%) | **Over 20 min** | Single dose; **max 2 g** | Monitor BP, reflexes. Medscape: 25–50 mg/kg over 10–20 min. Borrow from maternity, but dilute |
| **Aminophylline** | Load **5 mg/kg** (max 500 mg; UCG 2023: 5 mg/kg slow IV) | IV over **20 min** | Then **1 mg/kg/h** infusion (or 5 mg/kg 6-hourly over 20 min per WHO pocket book) | Stop when improving | **Omit the load if on oral theophylline/aminophylline.** Cardiac monitor; stop if vomiting, arrhythmia, seizure. UCG discourages because of poor safety |
| **Adrenaline** (if no salbutamol, or anaphylaxis) | **0.01 mg/kg = 0.01 mL/kg of 1:1000**, max 0.5 mg | IM (outer thigh) or SC (WHO pocket book for wheeze when no salbutamol) | Repeat after 5–15 min | – | First-line for **anaphylaxis**, not for routine asthma |
| **IV salbutamol** (HDU) | 15 µg/kg (max 250 µg) over 10 min, then 1–5 µg/kg/min | IV | Infusion | – | Specialist/HDU; monitor $K^+$, lactate, ECG |
| **Paracetamol** (if febrile) | 15 mg/kg | Oral | 6-hourly | Max 60 mg/kg/day | Avoid ibuprofen/aspirin if NSAID-sensitive |

**UGANDA CONTEXT**

UCG 2023 dosing differs slightly from international guides: **prednisolone 1 mg/kg** (GINA/BTS 1–2 mg/kg), **hydrocortisone 4 mg/kg (max 100 mg)**, **ipratropium 250 µg** in children <12 y, **aminophylline 5 mg/kg slow IV**, and **no magnesium** listed for children. Most Kampala referral wards (Mulago, Mengo, Nsambya) now use **IV $MgSO_4$ before aminophylline**, in line with GINA 2025 and BTS/SIGN. Quote UCG in the written test and the international sequence when asked "what else could you give?"

## D. Setting-specific plan

| Setting | What to do |
|---|---|
| **Home / VHT** | Salbutamol 2–4 puffs via spacer, repeat every 20 min up to 10 puffs; if not better within 1 hour or any danger sign, go to the nearest facility (action plan) |
| **HC III** (UCG level for mild–moderate) | Salbutamol by spacer or nebuliser every 20–30 min, prednisolone 1 mg/kg; observe 30–60 min. **Refer** if not improving, relapse within 3–4 h, or severe features |
| **HC IV / general hospital** (UCG level for severe) | Admit; oxygen; salbutamol + ipratropium; prednisolone or hydrocortisone; aminophylline (UCG) or **$MgSO_4$** (borrowed from maternity) if not responding; refer life-threatening asthma after first aid |

| Setting | What to do |
|---|---|
| **Regional / national referral** | All of the above; $MgSO_4$; HDU monitoring; blood gas; CXR; IV salbutamol; PICU/ventilation (Mulago and a few private hospitals) |

**Referral checklist**: oxygen during transfer (cylinder or concentrator with inverter), a health worker who can give salbutamol on the way, steroid given, referral note with times and doses, call ahead.

## E. Admitted child: ward care

- **Observations** every 15–30 min initially (RR, HR, $SpO_2$, work of breathing, talk/feed), then 1–4-hourly as improving.
- **Salbutamol** weaned stepwise: 20-minutely → hourly → 2-hourly → 4-hourly. Switch nebuliser → spacer as soon as possible (it trains technique).
- **Oxygen** weaned when $SpO_2$ ≥94% in air.
- **Fluids**: encourage oral fluids; if IV needed, **maintenance only** (Holliday–Segar: 100 mL/kg/day for the first 10 kg, 50 mL/kg/day for the next 10 kg, 20 mL/kg/day thereafter); avoid overload (high negative intrathoracic pressure promotes pulmonary oedema; SIADH can occur).
- **Antibiotics only** if fever with focal signs, CXR consolidation or clear bacterial infection.
- **Teach** inhaler + spacer technique on the ward, every day, to the parent and child.

## F. Discharge plan (the part that prevents the next attack)

**Ready for discharge when**:

- Stable on salbutamol **3–4-hourly** (not needing it more often),
- **$SpO_2$ ≥94% in air** (UCG flowchart: >94%),
- PEF **>60–80%** of best/predicted (if able),
- Eating, drinking and talking normally,
- Parent/child has demonstrated **correct inhaler and spacer technique**.

**Send home with**:

1. **Prednisolone** to complete **3 days** (UCG 2023 for children; up to 5 days).
2. **Salbutamol pMDI + spacer**: 2–4 puffs 4-hourly for 1–2 days, then as needed.
3. **Start or step up a controller**: e.g. **beclomethasone 100 µg twice daily** via spacer (UCG Step 2 for children: 100–200 µg 12-hourly), or ICS-formoterol in adolescents.
4. **Written asthma action plan** (green/yellow/red zones): what to do when well, when getting worse, and when to come straight to hospital.
5. **Trigger advice**: move cooking smoke out of the house or improve ventilation, no smoking indoors, damp-dust, treat colds early.
6. **Follow-up** within **2 days to 1 week** (UCG: review in 48 h for mild cases; 1–2 weeks for chronic plan), and at 1–3 months.
7. **School letter** to allow the inhaler at school and during sports.

## G. Chronic control: stepwise treatment

**GINA 2025** (current international standard):

| Step | Children ≤5 y | Children 6–11 y | Adolescents ≥12 y (Track 1, preferred) |
|---|---|---|---|
| 1 | As-needed SABA (consider short course of ICS at the start of viral illness) | Low-dose ICS taken **whenever SABA is taken** | **As-needed low-dose ICS-formoterol** (AIR) |

| Step | Children ≤5 y | Children 6–11 y | Adolescents ≥12 y (Track 1, preferred) |
|---|---|---|---|
| 2 | **Daily low-dose ICS** (preferred) | **Daily low-dose ICS** | As-needed low-dose ICS-formoterol |
| 3 | **Double** low-dose ICS | Low-dose ICS-LABA, or medium-dose ICS, or very-low-dose ICS-formoterol **MART** | **Low-dose MART** (ICS-formoterol maintenance and reliever) |
| 4 | Continue controller and **refer** | Medium-dose ICS-LABA or low-dose MART; refer | Medium-dose MART |
| 5 | – | Phenotype; add-on (e.g. tiotropium, biologics); specialist | Add-on LAMA; phenotype; biologics; specialist |

**Low daily ICS doses (GINA 2025)**: beclomethasone dipropionate (HFA/extrafine) **50–100 µg/day** (6–11 y), 100 µg/day (≤5 y); budesonide DPI 100–200 µg/day; fluticasone propionate 50–100 µg/day. **Always via a spacer in children.**

UGANDA CONTEXT

**UCG 2023 chronic asthma steps** (older format): Step 1 intermittent → salbutamol as needed; Step 2 mild persistent → **beclomethasone 100–200 µg every 12 h** (children) + salbutamol prn; Step 3 moderate persistent → beclomethasone 100–400 µg 12-hourly (5–12 y); **children <5 y refer to a specialist**; Step 4 severe → **refer** (especially children <12 y). If no inhaler: salbutamol tablets (2–5 y 1–2 mg per dose; <2 y 100 µg/kg) as a last resort. Rescue prednisolone course: <1 y 1–2 mg/kg; 1–5 y up to 20 mg; 5–15 y up to 40 mg daily for 3–5 days. ICS-formoterol (budesonide-formoterol) is available in Kampala private pharmacies and some referral clinics but is expensive; **beclomethasone + salbutamol inhalers are what most Ugandan families can get.** GINA 2025 no longer recommends **SABA-only** treatment for anyone ≥6 years, because it increases the risk of severe attacks.

**Every review**: check **symptom control, inhaler technique, adherence, triggers, growth** (ICS at low dose has a tiny effect on height, outweighed by the benefit), and step down after 3 months of good control.

# Complications

| Complication | Clues | Action |
|---|---|---|
| **Respiratory failure / arrest** | Exhaustion, silent chest, drowsiness, rising $CO_2$ | ICU, ventilation |
| **Pneumothorax / pneumomediastinum** | Sudden deterioration, unilateral reduced air entry, **subcutaneous emphysema** | CXR; needle decompression then chest drain if tension |
| **Atelectasis / lobar collapse** (mucus plug) | Focal reduced breath sounds, CXR collapse | Usually resolves; physiotherapy after the acute phase |
| **Secondary pneumonia** | Fever, focal signs | Antibiotics |
| **Drug toxicity** | Salbutamol: tachycardia, hypokalaemia, lactic acidosis; aminophylline: vomiting, arrhythmia, seizures; Mg: hypotension | Reduce or stop drug, check $K^+$ |
| **Dehydration** | Poor intake, fast breathing | Oral or IV maintenance fluids |
| **Long-term** | Missed school, poor growth (uncontrolled asthma or repeated oral steroids), fixed airflow limitation, psychosocial stress, death | Controller therapy, education |

# Ward-round & exam pearls

**Model case presentation (child):**

> "Brian is a 7-year-old boy from Kawempe, weighing 22 kg (weight-for-age Z-score −0.5, height-for-age Z-score 0), fully immunised by UNEPI including PCV and both measles-rubella doses, HIV-negative (tested last year), on a normal family diet. He is a known asthmatic diagnosed at 4 years, using only a salbutamol inhaler without a spacer; no inhaled steroid. He presented at 2 a.m. with one day of cough and worsening breathlessness after a cold; his mother cooks on a charcoal stove indoors. He has had two admissions this year, no ICU stay. On arrival he could say only words, was sitting forward, $SpO_2$ 89% in air, RR 38, HR 138, temperature 37.6 °C, with intercostal recession and widespread expiratory wheeze, equal air entry, no crackles; PEF 40% of predicted. Assessment: **acute severe asthma**, viral trigger, on a background of uncontrolled asthma. He received oxygen, three back-to-back nebulisers of salbutamol 5 mg with ipratropium 250 µg, and prednisolone 40 mg. At 1 hour he speaks in sentences, $SpO_2$ 95% on 2 L, RR 28, PEF 65%. Plan: salbutamol 2-hourly then 4-hourly by spacer, prednisolone for 3 days, wean oxygen, start beclomethasone 100 µg twice daily via a bottle spacer, teach technique, written action plan, smoke advice, review in clinic in 1 week."

## How to present this tutorial (Edith presents, Week 1 Tuesday, 08:00–09:30, Dr Mawanda)

**Your goal**: in about **40 minutes of presenting + 30 minutes of discussion**, leave every classmate able to **grade an attack, treat it for the first hour, and write a discharge plan**. Dr Mawanda will judge you on clarity, accuracy of doses, and whether you can handle questions.

### Suggested structure and timing

| Time | Segment | What to say / do |
|---|---|---|
| 0–3 min | **Hook case** | Read a 4-line vignette (e.g. Brian above, at 2 a.m., speaking in words, $SpO_2$ 89%). Ask: "What grade is this, and what is your first action?" Take 2 answers. Don't correct yet |
| 3–8 min | **Definition & burden** | GINA 2025 definition (variable symptoms + variable airflow limitation); acute exacerbation definition; one Uganda slide: triggers (charcoal smoke, viral colds), what is available at HC III vs RRH |
| 8–14 min | **Pathophysiology in one diagram** | Draw the airway cross-section: smooth muscle (→ salbutamol, ipratropium, Mg), oedema/inflammation (→ steroids), mucus. Show the $CO_2$ phases (low → normal → high) |
| 14–22 min | **Severity grading table** | The single most important slide. Put the UCG 2023 table up and walk the columns. Emphasise **silent chest** and **normal $CO_2$ = danger** |
| 22–32 min | **First-hour management algorithm** | Oxygen 94–98% → salbutamol every 20 min (spacer dose and nebuliser dose) → steroid in first hour → ipratropium if severe → IV $MgSO_4$ 40–50 mg/kg max 2 g → aminophylline → ICU. Then show the **drug table** with mg/kg and max doses. Highlight UCG vs GINA differences in one [UGANDA] box |
| 32–36 min | **Discharge & chronic control** | Discharge criteria, the 7-item discharge pack, GINA 2025 steps in one simplified table, the bottle spacer (bring one if you can and **demonstrate** it) |
| 36–40 min | **Back to the hook case** | Answer it fully; summarise 5 take-home messages |
| 40–70 min | **Questions and discussion** | Use the viva questions below as backup; ask the class two MCQs from the study set |

### Tips for presenting well

- **Open with the case and close with the same case.** It makes the talk feel complete.
- **One idea per slide or board section**; no paragraphs. Tables beat text. If there is no projector, draw the **severity table** and the **algorithm** on the board before 08:00.
- **Say doses out loud with units and maxima** ("prednisolone one to two milligrams per kilogram, maximum forty milligrams in a seven-year-old"). Tutors notice dose errors more than anything else.
- **Name your sources**: "UCG 2023 says… GINA 2025 adds…". Have the UCG page numbers (acute asthma pp. 331–336, chronic asthma pp. 337–339) ready.
- **Demonstrate** the spacer technique (5 breaths per puff, one puff at a time). A 30-second demonstration is remembered better than any slide.
- **Prepare a one-page handout**: the severity table, the first-hour algorithm, and the drug table.
- **If you don't know an answer**, say: "I'm not sure; I'll check the UCG and get back to you after the ward round." Then do it.
- **Rehearse once with a timer** the night before; most first-time presenters overrun by 15 minutes.

**Questions Dr Mawanda is likely to ask you (prepare these)**

1. *How do you distinguish asthma from bronchiolitis in a 14-month-old?* Age, first vs recurrent episode, crackles vs wheeze, atopy, response to salbutamol.
2. *Why is a silent chest dangerous?* No airflow; impending respiratory arrest.
3. *What is the dose and route of magnesium, and why dilute it?* 40–50 mg/kg IV over 20 min, max 2 g; 50% solution is hyperosmolar and irritant; dilute to ≤20%.
4. *Why don't we give antibiotics?* Exacerbations are mostly viral; antibiotics only for clear bacterial infection.
5. *What would you do differently at a HC III?* Spacer salbutamol, oral prednisolone, observe 1 hour, refer severe cases with oxygen if available.
6. *How does GINA 2025 differ from UCG 2023 for chronic asthma?* GINA: no SABA-only treatment ≥6 y; ICS whenever SABA used (6–11 Step 1); ICS-formoterol as reliever in ≥12; UCG still uses SABA-only Step 1 and beclomethasone at Step 2.

## Likely viva questions with answers

1. *Define asthma.* A chronic inflammatory airway disease with variable symptoms of wheeze, breathlessness, chest tightness and cough, and variable expiratory airflow limitation (GINA 2025).
2. *Name three features of life-threatening asthma.* Silent chest, cyanosis/$SpO_2$ <92%, poor respiratory effort, exhaustion, altered consciousness, hypotension, PEF <33%.
3. *Salbutamol dose by spacer and by nebuliser?* 2–10 puffs (100 µg each) via spacer; 2.5 mg (<5 y) or 2.5–5 mg (≥5 y) nebulised; every 20 min in the first hour.
4. *When do you give ipratropium?* Severe or life-threatening attacks, 250 µg every 20–30 min for the first 2 hours, with salbutamol.
5. *Side effects of salbutamol?* Tachycardia, tremor, hypokalaemia, lactic acidosis, hyperglycaemia.
6. *Why is aminophylline less favoured?* Narrow therapeutic index (arrhythmias, seizures, vomiting), interactions, little added benefit over inhaled therapy + steroids.
7. *What makes a child at high risk of dying from asthma?* Previous ICU/intubation, frequent admissions, overuse of SABA, no ICS, poor adherence, psychosocial problems.

## OSCE tips

- **Inhaler/spacer station**: shake, attach, seal lips (or mask over nose and mouth for <3–4 y), one puff, 5 tidal breaths (or one slow deep breath and hold 10 s for older children), wait 30 s, repeat; rinse mouth after ICS; clean spacer weekly, air-dry (reduces static).
- **PEF station**: stand, reset the needle, deep breath, lips sealed, short sharp blow, best of three, compare with chart for height.

- **Counselling station**: explain reliever vs preventer ("blue opens the airway now; brown/orange calms the swelling over weeks, take it every day even when well").

# Mnemonics & memory aids

- **Life-threatening asthma: "33, 92, CHEST"**: PEF **<33%**, $SpO_2$ **<92%**, **C**yanosis, **H**ypotension, **E**xhaustion/altered consciousness, **S**ilent chest, **T**ired (poor effort).
- **Treatment ladder: "O SHIMA"**: **O**xygen, **S**albutamol, **H**ydrocortisone/prednisolone, **I**pratropium, **M**agnesium, **A**minophylline (then ICU).
- **Discharge "S-P-A-C-E-R"**: **S**teroid course, **P**reventer inhaler, **A**ction plan, **C**heck technique, **E**nvironment/triggers, **R**eview date.
- **Salbutamol side effects: "TTT-L"**: **T**achycardia, **T**remor, **T**oo-little potassium, **L**actate.

# High-yield summary

## Numbers & doses to know

| Item | Value |
|---|---|
| Severe: HR | **>140 (2–5 y)**, **>125 (>5 y)** |
| Severe: RR | **>40 (2–5 y)**, **>30 (>5 y)** |
| Severe: $SpO_2$ / PEF | **<92%** / **33–50%** |
| Life-threatening PEF | **<33%** |
| Oxygen target | **$SpO_2$ 94–98%** (GINA 2025; UCG ≥94%) |
| Salbutamol pMDI | **2–10 puffs** (100 µg) via spacer every 20 min in the first hour |
| Salbutamol nebulised | **2.5 mg (<5 y)**; 2.5–5 mg (≥5 y) every 20 min |
| Ipratropium | **250 µg** neb (<12 y), every 20–30 min for the first 2 h |
| Prednisolone | **1–2 mg/kg** once daily, max **20/30/40 mg** (<2 / 2–5 / ≥6 y), **3 days** |
| Hydrocortisone | **4 mg/kg IV** 6-hourly, max **100 mg** |
| IV magnesium sulfate | **40–50 mg/kg over 20 min, max 2 g** |
| Aminophylline | **5 mg/kg IV over 20 min**, then **1 mg/kg/h**; no load if on theophylline |
| Adrenaline (anaphylaxis) | **0.01 mg/kg IM** (0.01 mL/kg of 1:1000), max 0.5 mg |
| Weight estimate (1–10 y) | **(age + 4) × 2 kg** |
| Discharge | Salbutamol **≥3–4-hourly**, $SpO_2$ **≥94% air**, PEF **>60–80%**, technique checked |
| Reversibility (spirometry) | $FEV_1$ rise **≥12%** after bronchodilator |
| Low-dose ICS (6–11 y) | Beclomethasone **50–100 µg/day** (GINA 2025); UCG 100–200 µg 12-hourly |

## Classic exam traps

- **"The wheeze has gone, so he's better."** Not if he is drowsy with a falling RR: **silent chest**.
- **"Give diazepam because the child is agitated."** Never. Agitation = hypoxia.
- **"Start amoxicillin for every admitted asthmatic."** No: antibiotics only with evidence of bacterial infection.
- **"Nebuliser is always better than a spacer."** No: spacer is as effective in mild–moderate attacks and teaches technique.

- **"Aminophylline loading dose for everyone."** No: omit the load if the child takes oral theophylline/aminophylline; and $MgSO_4$ comes first in current guidelines.
- **"Prednisolone must be tapered."** Not after a 3–5-day course.
- **"First wheeze at 6 months = asthma."** Usually bronchiolitis.
- **"A normal $pCO_2$ is reassuring."** In a breathless asthmatic it means tiring.

## Questions seniors ask

**What is your first action for a child with severe asthma?** Sit them up, oxygen to $SpO_2$ 94–98%, and start salbutamol (with ipratropium if severe) every 20 minutes while you grade the attack and weigh the child.

**When would you give magnesium, and how much?** For severe or life-threatening asthma not responding to an hour of inhaled salbutamol + ipratropium and a steroid: 40–50 mg/kg IV over 20 minutes, max 2 g, diluted, with BP monitoring.

**How do you know your treatment is working?** Re-grade every 20–30 min: talks in sentences, falling RR and work of breathing, $SpO_2$ rising, PEF rising; tachycardia alone can be a salbutamol effect.

**What would you send this child home with?** Prednisolone to complete 3 days, salbutamol with a spacer, an inhaled steroid, a written action plan, trigger advice, a school letter and a review in 2–7 days.

**Why does this child keep coming back?** Usually no controller, poor technique (no spacer), poor adherence, indoor smoke, or wrong diagnosis (TB, foreign body, heart disease). Check all five.

**How would you treat anaphylaxis presenting with wheeze?** IM adrenaline 0.01 mg/kg (max 0.5 mg) into the thigh, repeat in 5 minutes if needed, oxygen, fluids for shock, then salbutamol, steroid and antihistamine as add-ons.

**Which children with asthma need a specialist?** Under-5s needing more than Step 2 (UCG), any child at Step 4, life-threatening attacks, poor growth, doubt about the diagnosis.

SELF-TEST

# Flashcards

27 cards. Cover the right-hand column and test yourself.

| # | Question | Answer |
|---|---|---|
| 1 | Define asthma (GINA 2025). | A chronic inflammatory airway disease with **variable respiratory symptoms** (wheeze, breathlessness, chest tightness, cough) and **variable expiratory airflow limitation**. |
| 2 | Features of a SEVERE asthma attack in a child (UCG 2023/BTS). | Can't complete sentences or too breathless to talk/feed; **$SpO_2$ <92%**; **HR >140 (2–5 y) or >125 (>5 y)**; **RR >40 (2–5 y) or >30 (>5 y)**; accessory muscle use; **PEF 33–50%**. |
| 3 | Features of LIFE-THREATENING asthma. | **Silent chest**, cyanosis, $SpO_2$ <92%, poor respiratory effort, **exhaustion**, **drowsiness/confusion**, bradycardia or hypotension, **PEF <33%**, normal or rising $pCO_2$. |
| 4 | Oxygen target in acute asthma. | **$SpO_2$ 94–98%** (GINA 2025). UCG 2023: high-flow oxygen at least 5 L/min to keep $SpO_2$ ≥94%. Use oxygen to drive nebulisers in severe attacks. |
| 5 | Salbutamol doses in an acute attack (spacer and nebuliser). | pMDI 100 µg/puff: **2–10 puffs via spacer** (one puff at a time, 5 breaths each). Nebulised: **2.5 mg (<5 y)**, **2.5–5 mg (≥5 y)**. Every **20 min for the first hour**, then by response. |
| 6 | Ipratropium in acute asthma: who, dose, how often? | Severe or life-threatening attacks only. **250 µg nebulised** (<12 y; 500 µg ≥12 y) mixed with salbutamol, every **20–30 min for the first 2 h**, then 4–6-hourly (UCG 2023). |
| 7 | Oral prednisolone for acute asthma: dose, max, duration. | **1–2 mg/kg once daily** (UCG: 1 mg/kg). Max **20 mg (<2 y), 30 mg (2–5 y), 40 mg (6–11 y)**. **3 days** for children (UCG 2023), up to 5. No taper. |
| 8 | IV hydrocortisone dose in paediatric asthma. | **4 mg/kg IV every 6 h**, **max 100 mg** per dose (UCG 2023). Use when vomiting, drowsy or life-threatening; switch to oral prednisolone as soon as possible. |
| 9 | IV magnesium sulfate for severe asthma: dose and rate. | **40–50 mg/kg IV (max 2 g) over 20 minutes**, single dose, diluted. Monitor BP and reflexes. First IV bronchodilator in GINA 2025/BTS/RCH. |
| 10 | Aminophylline in acute asthma: dose and key caution. | Load **5 mg/kg IV over 20 min**, then **1 mg/kg/h**. **Omit the load if already on oral theophylline/aminophylline.** Narrow therapeutic window (vomiting, arrhythmias, seizures); UCG 2023 discourages its use. |
| 11 | Why is a silent chest ominous? | Airways are so narrow that almost **no air moves**, so there is no wheeze. With drowsiness and a falling RR it signals **exhaustion and impending respiratory arrest**. |
| 12 | What does a 'normal' $pCO_2$ mean in a breathless asthmatic child? | **Danger.** Early in an attack the child hyperventilates (low $pCO_2$). Normal or rising $pCO_2$ (>6 kPa / 45 mmHg) = **tiring/respiratory failure** → HDU/ICU. |
| 13 | Side effects of high-dose salbutamol. | **Tachycardia, tremor, hypokalaemia, lactic acidosis**, hyperglycaemia, agitation. Judge the asthma by work of breathing and $SpO_2$, not by HR alone. |
| 14 | Discharge criteria after an asthma attack. | Stable on salbutamol **3–4-hourly**; **$SpO_2$ ≥94% in air**; PEF **>60–80%** of best; eating and talking normally; **inhaler/spacer technique checked**. |

| # | Question | Answer |
|---|---|---|
| 15 | The discharge pack for an asthmatic child (S-P-A-C-E-R). | **S**teroid course to finish, **P**reventer (ICS) started/stepped up, written **A**ction plan, **C**heck technique (spacer), **E**nvironment/trigger advice, **R**eview in 2–7 days. |
| 16 | How do you make a spacer from a plastic bottle? | Take a clean **500 mL** water bottle, cut a hole in the base the shape of the inhaler mouthpiece, seal the gap with tape; the child breathes through the bottle neck. Works as well as a commercial spacer. |
| 17 | Spacer technique for a child. | Shake inhaler, fit to spacer, seal lips on the mouthpiece (face mask for <3–4 y), **one puff at a time**, **5 tidal breaths** per puff (or a slow deep breath and 10 s hold), wait 30 s, repeat. Rinse mouth after ICS. |
| 18 | GINA 2025 Step 1–2 for children 6–11 years. | Step 1: **low-dose ICS taken whenever SABA is taken**. Step 2: **daily low-dose ICS** (plus as-needed SABA). SABA-only treatment is no longer recommended for ≥6 y. |
| 19 | GINA 2025 preferred reliever for adolescents ≥12 years. | **As-needed low-dose ICS-formoterol** (anti-inflammatory reliever, AIR) at Steps 1–2; **MART** (ICS-formoterol maintenance and reliever) from Step 3. |
| 20 | GINA 2025 for children ≤5 years with frequent wheeze. | Step 2: **daily low-dose ICS** (e.g. beclomethasone 100 µg/day via spacer + mask); Step 3: double the low dose; Step 4: refer. Intermittent ICS at the start of viral illness is an alternative. |
| 21 | UCG 2023 chronic asthma Step 2 dose for children. | **Beclomethasone 100–200 µg every 12 h** + salbutamol 1–2 puffs prn. Children **<5 y needing Step 3, and any child needing Step 4: refer** to a specialist. |
| 22 | Risk factors for a fatal asthma attack. | Previous **ICU/intubation**; ≥2 admissions or ≥3 ER visits in a year; using >1 salbutamol canister/month; **no ICS**; recent oral steroids; poor adherence; psychosocial problems; food allergy. |
| 23 | Wheeze + urticaria + hypotension after eating groundnuts. First drug? | **IM adrenaline 0.01 mg/kg** (0.01 mL/kg of 1:1000), max 0.5 mg, outer thigh; repeat after 5 min if needed. This is **anaphylaxis**, not plain asthma. |
| 24 | Why never sedate a child with acute asthma? | Restlessness is usually **hypoxia**. Sedatives (diazepam, promethazine) depress respiratory drive and can precipitate **respiratory arrest**. |
| 25 | When is a chest X-ray indicated in acute asthma? | Not routinely. Only for **life-threatening** or non-responding attacks, suspected **pneumothorax** or surgical emphysema, **focal signs/fever** (pneumonia), suspected **foreign body**, or diagnostic doubt. |
| 26 | Estimate the weight of a 6-year-old with no scale. | **(age + 4) × 2 = (6 + 4) × 2 = 20 kg** (APLS formula for 1–10 y). Weigh as soon as possible; use a Broselow tape if available. |
| 27 | How does UCG 2023 differ from GINA 2025 in the acute attack? | UCG: prednisolone **1 mg/kg**, aminophylline as the IV step, **no magnesium** for children. GINA/BTS: prednisolone 1–2 mg/kg (capped), **IV $MgSO_4$ before aminophylline**. Kampala referral wards use $MgSO_4$ first. |

SELF-TEST

# Single best answer questions

15 questions. The answers and explanations follow at the end.

**1.** A 6-year-old girl (20 kg) with known asthma is brought to Mengo emergency at 3 a.m. She is sitting forward and can say only 2–3 words at a time. RR 36/min, HR 142/min, $SpO_2$ 90% in air, widespread wheeze, temperature 37.4 °C. Which grade of attack is this?

A Mild, because she is still wheezing
B Moderate, because $SpO_2$ is above 88%
C Severe
D Life-threatening, because RR is above 30
E Cannot be graded without a peak flow

**2.** The girl in the previous question (20 kg, severe asthma, $SpO_2$ 90%) has just been placed on oxygen. Which treatment should be given NEXT?

A Nebulised salbutamol 5 mg with ipratropium 250 µg, repeated every 20 minutes
B Oral salbutamol syrup 2 mg and observe for 1 hour
C IV aminophylline 5 mg/kg as the first bronchodilator
D Amoxicillin 40 mg/kg and paracetamol
E Diazepam 0.2 mg/kg IV to calm her so she can use the nebuliser

**3.** A 4-year-old boy weighing 15 kg is being treated for a moderate asthma attack at a HC IV. He has had salbutamol by spacer. He is not vomiting. What is the correct steroid prescription according to UCG 2023/GINA 2025?

A Prednisolone 60 mg once daily for 7 days, then taper
B Dexamethasone 0.6 mg/kg IM once only, because oral steroids are ineffective
C Hydrocortisone 100 mg IV 6-hourly for 5 days
D Prednisolone 15–30 mg orally once daily for 3 days
E No steroid, because steroids are only for severe attacks

**4.** A 10-year-old boy (30 kg) with severe asthma has had oxygen, three salbutamol + ipratropium nebulisers and IV hydrocortisone over the past hour. He is still speaking in words, $SpO_2$ 91% on 6 L/min oxygen, RR 34, widespread wheeze. Magnesium sulfate 50% is available from the maternity ward. What is the MOST appropriate next step?

A Magnesium sulfate 4 g IV over 15 minutes, the eclampsia loading dose
B Magnesium sulfate 1.2–1.5 g IV, diluted, over 20 minutes
C Stop nebulisers because he has had too much salbutamol
D Promethazine 0.5 mg/kg IM to settle him
E Discharge on oral prednisolone because the steroid will work in 6 hours

**5.** An 8-year-old girl has had hourly nebulised salbutamol for 4 hours. She is now drowsy, RR has fallen from 40 to 20/min, the chest is very quiet on auscultation and $SpO_2$ is 88% on a face mask. Her mother says she is 'calmer now'. What is the correct interpretation?

A She is improving; reduce salbutamol to 4-hourly
B She has salbutamol toxicity; stop all bronchodilators
C She has fallen asleep from exhaustion after a long night, which is reassuring
D She has developed pneumonia; start ceftriaxone and continue the same plan
E She has life-threatening asthma with impending respiratory arrest

**6.** A 3-year-old boy (14 kg) presents with sudden onset of cough and wheeze that started while he was eating groundnuts with his siblings 2 hours ago. He has no previous wheeze and no fever. Breath sounds are reduced over the right lower zone with a localised wheeze. $SpO_2$ 95%. What is the MOST likely diagnosis?

A Inhaled foreign body
B First episode of asthma
C Bronchiolitis
D Lobar pneumonia
E Anaphylaxis to groundnuts

**7.** A 12-year-old boy (35 kg) with asthma presents with wheeze, generalised urticaria, lip swelling and BP 70/40 mmHg 15 minutes after a bee sting. Which is the FIRST drug to give?

A Nebulised salbutamol 5 mg
B Hydrocortisone 100 mg IV
C Adrenaline 0.35 mg (0.35 mL of 1:1000) IM into the thigh
D Chlorphenamine 4 mg orally
E Aminophylline 5 mg/kg IV

**8.** A 7-year-old girl was admitted with severe asthma and is now on salbutamol by spacer 4-hourly. $SpO_2$ is 96% in air, she is eating and running around the ward. Her only home treatment has been salbutamol tablets bought from a drug shop. Which discharge plan is MOST appropriate?

A Continue salbutamol tablets 2 mg three times daily and review if she wheezes again
B Oral prednisolone for 1 month to prevent relapse
C Salbutamol inhaler without a spacer, as needed
D Amoxicillin for 5 days to prevent a chest infection, and salbutamol tablets
E Finish prednisolone, salbutamol inhaler with a spacer, start beclomethasone via spacer, written action plan and review in 1 week

**9.** A 14-month-old girl presents in the rainy season with 2 days of runny nose, then cough, fast breathing and difficulty feeding. RR 58/min, $SpO_2$ 93%, bilateral fine crackles and some wheeze. This is her first episode. A trial of salbutamol produces no change. Which diagnosis is MOST likely?

A Acute severe asthma
B Bronchiolitis
C Inhaled foreign body
D Heart failure from a VSD
E Pertussis

**10.** A 9-year-old boy with severe asthma has received IV magnesium. He is started on IV aminophylline. Which history item would make you OMIT the loading dose? Advanced

A He received hydrocortisone 1 hour ago
B He is HIV-positive on dolutegravir-based ART
C He has had three salbutamol nebulisers
D He has been taking oral aminophylline tablets at home
E He is allergic to penicillin

**11.** After 3 hours of frequent nebulised salbutamol, a 6-year-old with improving asthma has HR 165/min, a fine tremor, and serum potassium 2.9 mmol/L. She now speaks in sentences and $SpO_2$ is 96% on 1 L/min. What is the BEST explanation and response? Advanced

- A Worsening asthma; give IV aminophylline
- B Sepsis; take blood cultures and give ceftriaxone
- C Expected $\beta_2$-agonist effects; space salbutamol out and correct potassium
- D Supraventricular tachycardia; give adenosine
- E Anxiety; give oral diazepam

**12.** A 10-year-old girl with asthma uses her salbutamol inhaler 4–5 times a week and wakes at night coughing twice a week. She has no inhaled steroid. According to GINA 2025, what is the preferred controller?

- A Daily low-dose inhaled corticosteroid via a spacer
- B Salbutamol tablets 2 mg three times daily
- C Oral prednisolone 10 mg daily
- D Montelukast alone as first choice
- E Increase salbutamol to regular 4-hourly use

**13.** Which ONE of the following is a recognised feature of life-threatening (rather than severe) asthma in a 7-year-old?

- A Heart rate 130/min
- B Respiratory rate 34/min
- C Use of accessory muscles
- D Altered level of consciousness
- E Peak flow 45% of predicted

**14.** A 5-year-old (18 kg) with severe asthma is vomiting every oral dose. You need to give a systemic steroid. What is the MOST appropriate prescription?

- A Prednisolone 18 mg orally; repeat if vomited
- B Hydrocortisone 72 mg IV every 6 hours until oral prednisolone is tolerated
- C Dexamethasone 16 mg IV every 6 hours
- D Beclomethasone 400 µg inhaled as a single high dose instead
- E Methylprednisolone 30 mg/kg IV daily for 3 days

**15.** At a HC III with no oxygen, a 9-year-old girl (25 kg) with moderate asthma has received salbutamol 6 puffs via a bottle spacer three times over an hour and prednisolone 25 mg. She now speaks in sentences, RR 24/min, no recession. What should you do?

- A Refer immediately to the regional referral hospital
- B Give aminophylline 5 mg/kg IV at the HC III
- C Admit at the HC III for 5 days of IV hydrocortisone
- D Discharge with salbutamol tablets only
- E Observe for a further 1–3 hours; if stable, discharge with prednisolone to complete 3 days, inhaler + spacer, ICS, action plan and review in 48 hours

## Answers & explanations

**1. C** · Severe

C is right: inability to complete sentences, $SpO_2$ <92%, HR >125 and RR >30 in a child >5 years are all **severe** features. A is wrong: wheeze is present in every grade except the silent chest; it does not mean mild. B is wrong: the cut-off is 92%, not 88%. D is wrong: RR >30 is a severe, not life-threatening, feature; she has no silent chest, cyanosis, exhaustion or altered consciousness. E is wrong: grading is clinical; PEF helps but is not required (and is unreliable in a breathless 6-year-old).

**2. A** · Nebulised salbutamol 5 mg with ipratropium 250 µg, repeated every 20 minutes

A is right: in a severe attack the first-line treatment is inhaled **salbutamol plus ipratropium**, oxygen-driven, every 20 minutes for the first hour (UCG 2023, GINA 2025). B is wrong: oral salbutamol is slow, weak and causes side effects; it is not acute treatment. C is wrong: aminophylline is a later step after inhaled therapy, steroid (and magnesium) have failed. D is wrong: there is no evidence of bacterial infection; antibiotics are not routine. E is wrong: sedation in asthma can cause respiratory arrest; agitation is hypoxia.

**3. D** · Prednisolone 15–30 mg orally once daily for 3 days

D is right: prednisolone **1–2 mg/kg** (15–30 mg for 15 kg; max 30 mg at 2–5 years) once daily for **3 days** is the paediatric course. A is wrong: 60 mg is far above the maximum for a 4-year-old, 7 days is unnecessarily long, and no taper is needed for short courses. B is wrong: oral steroids work as well as injected ones when the child can swallow. C is wrong: hydrocortisone is for children who cannot take oral medication, and the dose is 4 mg/kg (60 mg), not 100 mg. E is wrong: any attack needing more than a single dose of salbutamol should get a steroid early.

**4. B** · Magnesium sulfate 1.2–1.5 g IV, diluted, over 20 minutes

B is right: IV $MgSO_4$ **40–50 mg/kg** (30 kg → 1.2–1.5 g; max 2 g), diluted, over 20 minutes is the first IV bronchodilator for severe asthma not responding to an hour of inhaled therapy (GINA 2025, BTS/SIGN, RCH). A is wrong: the 4 g eclampsia load is double the paediatric maximum and could cause hypotension and respiratory depression. C is wrong: he is still severe; inhaled salbutamol should continue (1-hourly or back-to-back). D is wrong: sedation is dangerous in asthma. E is wrong: he is hypoxic and severe; discharge would be unsafe.

**5. E** · She has life-threatening asthma with impending respiratory arrest

E is right: drowsiness, a falling RR, a **silent chest** and low $SpO_2$ are classic signs of **exhaustion and impending arrest**: call senior help, back-to-back nebulisers, IV magnesium, prepare to ventilate. A is wrong: a falling RR without better $SpO_2$ and alertness is deterioration, not improvement. B is wrong: salbutamol toxicity causes tachycardia and tremor, not a silent chest; bronchodilators must continue. C is wrong: altered consciousness in asthma is a life-threatening feature. D is wrong: nothing suggests pneumonia, and antibiotics would not address the immediate threat.

**6. A** · Inhaled foreign body

A is right: **sudden onset while eating**, no previous wheeze, and **unilateral** reduced air entry with localised wheeze = **foreign body** (usually right main bronchus) until proven otherwise; get inspiratory and expiratory CXR and refer for bronchoscopy. B is wrong: asthma gives bilateral, diffuse signs and usually a history of recurrent episodes. C is wrong: bronchiolitis affects infants <2 years after a coryzal prodrome, with bilateral crackles/wheeze. D is wrong: there is no fever and the onset was sudden. E is wrong: anaphylaxis would have urticaria, angioedema, vomiting or hypotension and bilateral wheeze.

**7. C** · Adrenaline 0.35 mg (0.35 mL of 1:1000) IM into the thigh

C is right: this is **anaphylaxis** (skin + respiratory + hypotension after a sting). First drug: **IM adrenaline 0.01 mg/kg** (35 kg → 0.35 mg; max 0.5 mg) into the outer thigh, repeated after 5 minutes if needed. A is wrong: salbutamol helps bronchospasm but does nothing for the shock and angioedema; it is an add-on. B is wrong: steroids act over hours and are adjuncts. D is wrong: antihistamines treat itch only and never replace adrenaline. E is wrong: aminophylline has no role in anaphylaxis.

**8. E** · Finish prednisolone, salbutamol inhaler with a spacer, start beclomethasone via spacer, written action plan and review in 1 week

E is right: she meets discharge criteria; the plan must **finish the steroid course**, provide an inhaled reliever **with a spacer**, **start a controller** (ICS; she has had a severe attack, so Step 2 or above), a written action plan, trigger advice and early follow-up. A is wrong: tablets are slow and side-effect-prone and leave the inflammation untreated. B is wrong: long courses of oral steroids cause growth suppression and other harm; inhaled steroids are the controller. C is wrong: without a spacer a child gets very little drug, and SABA-only therapy is not recommended for ≥6 y (GINA 2025). D is wrong: prophylactic antibiotics have no role.

**9. B** · Bronchiolitis

B is right: a **first** episode in a child <2 years after a **coryzal prodrome**, with **fine crackles** and wheeze and **no response to salbutamol**, is bronchiolitis (usually RSV). A is wrong: asthma is rarely diagnosed at this age after a first episode, and crackles are not typical. C is wrong: a foreign body causes sudden onset and unilateral signs. D is wrong: there is no murmur, hepatomegaly or chronic poor feeding/growth history. E is wrong: pertussis causes paroxysmal cough with whoop or apnoea, usually with a clear chest between paroxysms.

**10. D** · He has been taking oral aminophylline tablets at home

D is right: if a child already takes **oral theophylline/aminophylline**, a loading dose risks **toxic levels** (vomiting, arrhythmias, seizures); start the maintenance infusion only, ideally guided by a level. A is wrong: steroids do not interact with aminophylline in this way. B is wrong: dolutegravir does not mandate omitting the load (rifampicin and some PIs alter theophylline levels, but not DTG). C is wrong: salbutamol does not change the loading decision, though both drugs can lower potassium. E is wrong: penicillin allergy is irrelevant to aminophylline dosing.

**11. C** · Expected $\beta_2$-agonist effects; space salbutamol out and correct potassium

C is right: tachycardia, tremor and **hypokalaemia** are predictable **salbutamol effects**; her asthma is improving (talking, $SpO_2$ 96%), so space doses out and add potassium to fluids. A is wrong: her work of breathing and $SpO_2$ show improvement, and aminophylline would worsen tachycardia and hypokalaemia. B is wrong: there is no fever or other sign of sepsis. D is wrong: sinus tachycardia of 165 with a clear cause is not SVT (usually >220 in children, fixed rate). E is wrong: sedation is contraindicated in acute asthma.

**12. A** · Daily low-dose inhaled corticosteroid via a spacer

A is right: symptoms more than twice a week with night waking need a **daily low-dose ICS** (Step 2–3 for 6–11 years) with a spacer; UCG 2023 Step 2 is beclomethasone 100–200 µg 12-hourly. B is wrong: oral $\beta_2$-agonists are not controllers. C is wrong: daily oral steroids are reserved for the most severe asthma under specialist care. D is wrong: montelukast is a less effective alternative, not first choice. E is wrong: regular SABA without ICS increases the risk of severe attacks.

**13. D** · Altered level of consciousness

D is right: **altered consciousness** (drowsiness, confusion) is a life-threatening feature. A is wrong: HR >125 in a child >5 y is a severe feature. B is wrong: RR >30 is severe. C is wrong: accessory muscle use is severe. E is wrong: PEF 33–50% is severe; life-threatening is <33%.

**14. B** · Hydrocortisone 72 mg IV every 6 hours until oral prednisolone is tolerated

B is right: **hydrocortisone 4 mg/kg IV** (18 kg → 72 mg; max 100 mg) every 6 hours, then oral prednisolone once tolerated (UCG 2023). A is wrong: if she is vomiting every dose, the oral route is unreliable. C is wrong: 16 mg 6-hourly is a large overdose for an 18 kg child. D is wrong: inhaled steroids do not replace systemic steroids in a severe attack. E is wrong: 30 mg/kg is a pulse dose used in conditions such as lupus or transplant rejection, not asthma.

**15. E** · Observe for a further 1–3 hours; if stable, discharge with prednisolone to complete 3 days, inhaler + spacer, ICS, action plan and review in 48 hours

E is right: she has responded well. UCG 2023 says monitor for 30–60 min (or longer), then send home with prednisolone (3 days in children), start or step up chronic treatment, teach self-management and **review in 48 hours**. A is wrong: referral is for poor response, relapse within 3–4 h, or severe features. B is wrong: aminophylline is not needed in a responding child and is not a HC III drug. C is wrong: she is not severe and can take oral medication. D is wrong: tablets alone ignore both the reliever technique and the controller.

SELF-TEST

# Clinical cases

Work through each step before reading the model answer.

## Case 1: Night-time severe asthma in a 7-year-old at Mengo

Brian, a 7-year-old boy from Kawempe, is carried into Mengo Hospital paediatric emergency at 2 a.m. by his mother. He has had a cold for 2 days and became very breathless tonight. He is sitting forward, can say only single words, and looks frightened. He is known to have asthma and uses a salbutamol inhaler 'when it is bad', without a spacer.

**1. What do you do in the first 2 minutes?**

> Move him to the resuscitation area, **sit him up** on his mother's lap, start **oxygen** (face mask 5–10 L/min) and attach a pulse oximeter. While a colleague prepares salbutamol, **grade the attack**: talk/feed, $SpO_2$, RR, HR, chest sounds, consciousness. Weigh him (or estimate (7 + 4) × 2 = 22 kg).

**2. Observations: weight 22 kg, $SpO_2$ 89% in air, RR 38, HR 138, T 37.6 °C, widespread wheeze, equal air entry, intercostal recession, alert. PEF 40%. Grade the attack.**

> **Acute severe asthma**: speaks in words, $SpO_2$ <92%, RR >30, HR >125, PEF 33–50%. Not life-threatening: alert, wheeze present (not silent), no cyanosis, PEF >33%.

**3. Write the first-hour treatment orders.**

> 1. Oxygen to $SpO_2$ **94–98%**.
> 2. **Salbutamol 5 mg + ipratropium 250 µg** nebulised with oxygen at 6–8 L/min, **every 20 min × 3**.
> 3. **Prednisolone 40 mg** orally now (1–2 mg/kg, max 40 mg at ≥6 y) or hydrocortisone 88 mg IV if vomiting.
> 4. Observations every 15 min; re-grade at 20, 40, 60 min.
> 5. No sedatives, no antibiotics (no focal signs).

**4. History taken during the nebulisers: two admissions this year, never in ICU, no inhaled steroid, charcoal stove indoors, father smokes, fully immunised, HIV test negative last year, WAZ –0.5. What does this add?**

> He has **uncontrolled asthma** with risk factors for a severe attack (**no ICS**, repeated admissions, smoke exposure). It does not change the immediate plan but drives the discharge plan: controller, spacer, smoke advice, action plan. No ICU history is reassuring but does not lower vigilance.

**5. At 60 min: speaks in sentences, $SpO_2$ 95% on 2 L, RR 28, HR 150 with tremor, PEF 65%. Interpret and plan.**

> **Good response**; the higher HR and tremor are **salbutamol effects**. Admit for observation: salbutamol (spacer) 1–2-hourly then 4-hourly, ipratropium can stop after 2–6 h, continue prednisolone daily for 3 days, wean oxygen to keep ≥94%, check $K^+$ if salbutamol stays frequent.

**6. Next morning he is on salbutamol 4-hourly, $SpO_2$ 96% in air, eating well. Write his discharge plan.**

> **S-P-A-C-E-R**: finish **prednisolone** (2 more days); salbutamol inhaler 2–4 puffs 4-hourly for 1–2 days then as needed, **with a bottle spacer**; start **beclomethasone 100 µg twice daily** via spacer (UCG Step 2); **written action plan**; **environment** (cook outside or ventilate, father not to smoke indoors); technique checked by teach-back; **review in 1 week** and school letter.

**Take-away:** Grade in 2 minutes, treat by grade every 20 minutes, give the steroid in the first hour, and use the admission to fix control: spacer, inhaled steroid, smoke and an action plan.

## Case 2: The quiet chest: life-threatening asthma

Aisha, a 10-year-old girl weighing 30 kg, has been on the paediatric ward for 3 hours with severe asthma. She has had five salbutamol + ipratropium nebulisers and IV hydrocortisone. The night nurse calls you because 'she has calmed down and is sleeping, but the saturation machine keeps alarming'.

**1. What do you expect to find, and what are you worried about?**

> Worry about **exhaustion**: a child who 'calms down' with falling $SpO_2$ may be tiring. Look for **drowsiness, a falling RR, a silent chest, cyanosis, bradycardia** and poor effort; these are life-threatening features.

**2. Findings: rousable only to voice, RR 18, very quiet chest bilaterally, $SpO_2$ 86% on 10 L mask, HR 150, BP 95/60. What is your diagnosis and first actions?**

> **Life-threatening asthma** with impending respiratory arrest. Call the **most senior clinician and anaesthetist** now. High-flow oxygen; **back-to-back salbutamol + ipratropium**; bag-valve-mask ready; second IV line; continue hydrocortisone.

**3. Which IV drug would you give next, and at what dose?**

> **Magnesium sulfate 40–50 mg/kg IV = 1.2–1.5 g** (max 2 g), diluted to ≤20% (e.g. 3 mL of 50% in 0.9% saline), **over 20 minutes**; monitor BP and reflexes. Borrow from maternity if needed, but never give the 4 g eclampsia dose.

**4. A capillary gas shows pH 7.28, $pCO_2$ 6.8 kPa (51 mmHg), lactate 3.8 mmol/L. Interpret.**

> **Respiratory acidosis** with a raised $pCO_2$ in a breathless asthmatic = **respiratory failure** from exhaustion. The lactate is partly from salbutamol. She needs HDU/ICU and probably ventilation.

**5. Twenty minutes after magnesium she is still drowsy with $pCO_2$ rising. What next?**

> **Aminophylline 5 mg/kg (150 mg) IV over 20 min** (if not on oral theophylline), then 1 mg/kg/h with cardiac monitoring, or IV salbutamol in HDU. Prepare for **intubation** by the most experienced person (ketamine induction; slow ventilation rate, long expiratory time). Arrange **PICU** (Mulago or private) and transfer with oxygen and an escort if not available on site.

**6. She recovers after 2 days in ICU. What must the discharge and follow-up include?**

> She is now **high risk for fatal asthma** (ICU admission): specialist asthma clinic, **daily ICS (or ICS-LABA/MART per GINA 2025)**, spacer, written action plan with clear 'go to hospital now' criteria, adherence support, trigger control, and early review (within 1 week).

**Take-away:** A calm, drowsy asthmatic with a quiet chest and falling $SpO_2$ is dying, not improving: call for help, magnesium, then aminophylline/ICU.

## Case 3: Recurrent wheeze in a toddler at a HC IV

Joel, a 3-year-old boy weighing 12 kg, is brought to a HC IV in Mukono with cough and wheeze for 1 day. His mother says it is the 'fourth time this year'; each time he gets salbutamol syrup. Between episodes he coughs at night and when he runs. His father has hay fever.

**1. What are your differentials and what in the history points to asthma?**

> Preschool **multiple-trigger wheeze/asthma** (recurrent episodes, night and exercise cough, atopic family) vs **episodic viral wheeze**; also consider TB (contact? weight loss?), foreign body (sudden onset?), HIV-related lung disease, heart failure, recurrent aspiration. Symptoms **between** colds and **atopy** favour asthma.

**2. Examination: alert, talking in phrases, RR 38, HR 130, $SpO_2$ 94% in air, mild subcostal recession, bilateral wheeze, no crackles, afebrile. WAZ −1.2, BCG scar present. Grade and treat.**

> **Mild–moderate** attack ($SpO_2$ ≥92%, RR ≤40, HR ≤140, talking). Give **salbutamol 2–6 puffs via spacer with a face mask** (or 2.5 mg nebulised) every 20 min for an hour, and **prednisolone 12–24 mg** (1–2 mg/kg; max 30 mg at 2–5 y) orally. Re-grade after each dose.

**3. He improves after 2 rounds. What should you check before deciding on chronic treatment?**

> **Growth** (WAZ −1.2: monitor), **TB exposure** and HIV status (offer a test if unknown), **immunisation** card (UNEPI: PCV, measles-rubella), feeding, **home smoke**, and whether the family can afford/use an inhaler with a spacer and mask.

**4. What controller does GINA 2025 recommend for him, and what does UCG 2023 say?**

> GINA 2025 (≤5 y): frequent episodes (≥3 per year) and interval symptoms → **daily low-dose ICS** (e.g. beclomethasone 100 µg/day via spacer + face mask), reviewed after 2–3 months. UCG 2023: Step 2 beclomethasone 100–200 µg 12-hourly; if a child **<5 y needs Step 3 or more, refer** to a specialist.

**5. How do you teach the mother to use the inhaler, and what do you tell her about salbutamol syrup?**

> Shake, attach to a spacer with a **face mask**, mask sealed over nose and mouth, **one puff, count 5–6 breaths**, repeat; rinse mouth/wipe face after ICS; clean the spacer weekly. Explain that **syrup is slow and causes shaking and fast heartbeat**; the inhaler with a spacer is quicker, safer and cheaper per dose. Give a written plan and review in 2–4 weeks.

**Take-away:** Recurrent wheeze with interval symptoms in an atopic toddler deserves an inhaled steroid via a spacer and mask, not repeated salbutamol syrup; check growth, TB and HIV along the way.

# References


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A study aid, not a protocol. Check doses against the Uganda Clinical Guidelines 2023, your unit's protocol and a senior before you treat a patient.